1.

Exercise 2

Define an MLP for regression, having 3 inputs, one hidden layer of 2 units, with the ReLU activation function, and 1 output. For the input $\boldsymbol{X} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$, if the weight matrices and bias vectors of the layers are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the output $\boldsymbol{O}$ of the network. Specify the total number of real parameters (weights and biases) for the network.

Solution:

we have the following formulas:

$$\begin{aligned} \boldsymbol{H} &= \sigma(\boldsymbol{X}\boldsymbol{W}^{(1)} + \boldsymbol{b}^{(1)}), \\ \boldsymbol{O} &= \boldsymbol{H}\boldsymbol{W}^{(2)} + \boldsymbol{b}^{(2)}, \end{aligned}$$

where $\boldsymbol{X} \in \mathbb{R}^{n \times d}$, $\boldsymbol{W}^{(1)} \in \mathbb{R}^{d \times h}$, $\boldsymbol{b}^{(1)} \in \mathbb{R}^{1 \times h}$, $\boldsymbol{H} \in \mathbb{R}^{n \times h}$, $\boldsymbol{W}^{(2)} \in \mathbb{R}^{h \times q}$, $\boldsymbol{b}^{(2)} \in \mathbb{R}^{1 \times q}$, $\boldsymbol{O} \in \mathbb{R}^{n \times q}$, and

$$\sigma(x) = \text{ReLU}(x) = \max(x, 0).$$

$n = 2$ (exemple), $d = 3$ (inputs), $h = 2$ (hidden units), $q = 1$ (output)

total parameters = $\underbrace{3 \times 2}_{W^{(1)}} + \underbrace{1 \times 2}_{b^{(1)}} + \underbrace{2 \times 1}_{W^{(2)}} + \underbrace{1 \times 1}_{b^{(2)}}$ ($d \times h$, $1 \times h$, $h \times q$, $1 \times q$)

$$W^{(1)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{pmatrix} \in \mathbb{R}^{3 \times 2} \qquad b^{(1)} = \begin{pmatrix} 1 & 0 \end{pmatrix} \in \mathbb{R}^{1 \times 2}$$

$$W^{(2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{R}^{2 \times 1} \qquad b^{(2)} = (1) \in \mathbb{R}^{(1)}$$

$$X W^{(1)} + b^{(1)} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 2 & -1 \end{pmatrix}$$

→ Broadcasting

$$H = \begin{pmatrix} \text{relu}(3) & \text{relu}(-1) \\ \text{relu}(2) & \text{relu}(-1) \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 2 & 0 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

$$O = H\,W^{(2)} + b^{(2)} = \begin{pmatrix} 3 & 0 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + (1) =$$

$$= \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \in^{2\times 1}$$

(2.) Problemă curs

$$x^{(1)} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad x^{(2)} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \quad x^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$y^{(1)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad y^{(2)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad y^{(3)} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$\bar{V}(x) = \text{ReLu}(x) = \max(0, x)$

$n = 3$, $d = 3$, $q = 2$

3 exemplii, 3 linii X, features, 2 linii Y

Presupunem $h = 2$

$$W^{(1)}_{\in \mathbb{R}^{d \times h}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \in \mathbb{R}^{3 \times 2} \qquad b^{(1)}_{\in \mathbb{R}^{1 \times h}} = \begin{pmatrix} 0 & 1 \end{pmatrix} \in \mathbb{R}^{1 \times 2}$$

$$W^{(2)}_{\in \mathbb{R}^{h \times q}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2 \times 2} \qquad b^{(2)}_{\in \mathbb{R}^{1 \times q}} = \begin{pmatrix} -1 & 1 \end{pmatrix} \in \mathbb{R}^{1 \times 2}$$

$$X = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3} \quad \rightarrow x^{(1)T}$$

nr parametrii = $3 \times 2 + 1 \times 2 + 2 \times 2 + 1 \times 2 = 14$

$$A = X W^{(1)} + b^{(1)} = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -1 & 2 \\ 2 & -1 \\ 1 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 2 & 0 \\ -1 & 0 \end{pmatrix}$$

$$H = \text{ReLu}(A) = \begin{pmatrix} 0 & 3 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

$$O = H W^{(2)} + b^{(2)} = \begin{pmatrix} 0 & 3 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ -1 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 1 \\ -1 & 3 \\ -1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

$$\mathcal{L}(w) = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{2} (\hat{y}^{(i)} - y)^2$$

$$= \frac{1}{2m} \text{Tr} \left[(O-Y)^T (O-Y) \right]$$

$Y^{(1)^T}$

$$O - Y = \begin{pmatrix} -4 & 1 \\ -1 & 3 \\ -1 & 1 \end{pmatrix} - \begin{pmatrix} \boxed{0 \quad 1} \\ 1 \quad 2 \\ 2 \quad 3 \end{pmatrix} = \begin{pmatrix} -4 & 0 \\ -2 & 1 \\ -3 & -2 \end{pmatrix}$$

$$\mathcal{L}(W^{(1)}, b^{(1)}, W^{(2)}, b^{(2)}) = \frac{1}{6} \operatorname{Tr}\left(\begin{pmatrix} -4 & -2 & -3 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} -4 & 0 \\ -2 & 1 \\ -3 & -2 \end{pmatrix} \right) =$$

$$= \frac{1}{6} \operatorname{Tr}\begin{pmatrix} 29 & 4 \\ 4 & 5 \end{pmatrix} = \frac{34}{6} = \frac{17}{3}$$

$L = \frac{1}{2m} \operatorname{Tr}\left((O-Y)^T (O-Y)\right)$ → pt ex 10

$R = \frac{\lambda}{2} \left(\|W^{(1)}\|_F^2 + \|W^{(2)}\|_F^2 \right)$ $\quad \|W\|_F = \sqrt{\operatorname{Tr}(W^T W)}$

$J = L + R$

$$\frac{\partial J}{\partial W^{(2)}} = \frac{\partial (L+R)}{\partial W^{(2)}} = \underbrace{\frac{\partial L}{\partial W^{(2)}}}_{\text{I}} + \underbrace{\frac{\partial R}{\partial W^{(2)}}}_{\text{II}}$$

I. $$\frac{\partial L}{\partial W^{(2)}} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial \theta}{\partial W^{(2)}} = \frac{\partial L}{\partial \theta} \frac{\partial (W^{(2)} h)}{\partial W^{(2)}} = \frac{\partial L}{\partial \theta} h^T$$

$h^T \cdot \frac{\partial L}{\partial \theta}$

$2 \times 3 \quad 3 \times 2$

- $$\frac{\partial L}{\partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{1}{2m} \operatorname{Tr}\left((O-Y)^T (O-Y)\right) \right) = \frac{1}{m} (O-Y)$$

$$\Rightarrow \frac{\partial L}{\partial W^{(2)}} = \frac{1}{m} h^T (O-Y)$$

II. $$\frac{\partial R}{\partial W^{(2)}} = \lambda W^{(2)}$$

$$\Rightarrow \frac{\partial J}{\partial W^{(2)}} = \frac{1}{m} H^T (O-Y) + \lambda W^{(2)} \in \mathbb{R}^{h \times 2}$$

$$\frac{\partial J}{\partial w^{(1)}} = \frac{\partial (L+s)}{\partial w^{(1)}} = \underbrace{\frac{\partial L}{\partial w^{(1)}}}_{\text{I}} + \frac{\partial s}{\partial w^{(1)}}$$

(I) $$\frac{\partial L}{\partial w^{(1)}} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w^{(1)}} = \frac{\partial L}{\partial z} \frac{\partial (w^{(1)} x)}{\partial w^{(1)}} = x^T \frac{\partial L}{\partial z} \in \mathbb{R}^{d \times h}$$

$d \times m \quad m \times h$

- $$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial z} = \frac{\partial L}{\partial h} \cdot \frac{\partial \phi(z)}{\partial z} = \frac{\partial L}{\partial h} \odot \phi'(z)$$

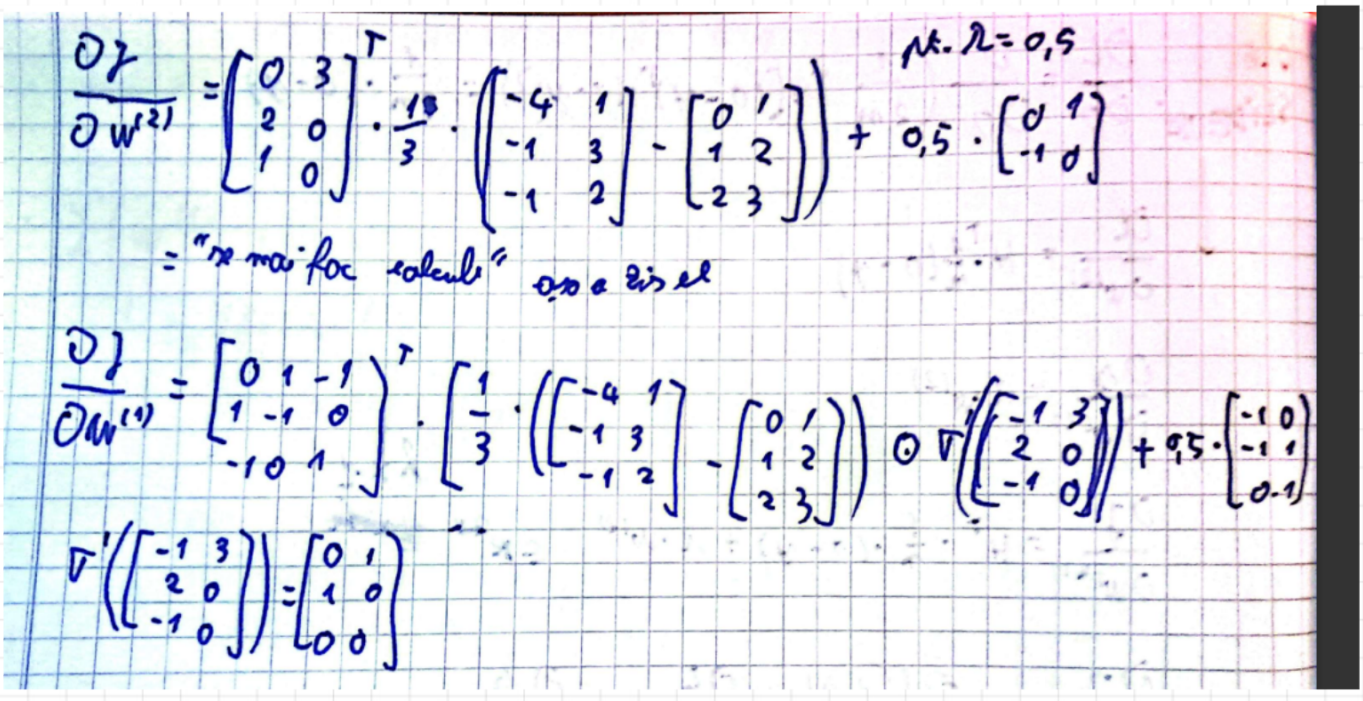
$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial \theta}{\partial h} = \frac{\partial L}{\partial \theta} \cdot \frac{\partial (w^{(2)} h)}{\partial h} = \frac{\partial L}{\partial \theta} w^{(2)T} \in \mathbb{R}^{m \times h}$$

$m \times q \quad q \times h$

$$= \frac{1}{m} (\theta - y) w^{(2)T}$$

(II) $$\frac{\partial s}{\partial w^{(1)}} = \lambda w^{(1)}$$

$$\frac{\partial J}{\partial w^{(1)}} = x^T \cdot \left(\frac{1}{m} (\theta - y) w^{(2)T} \right) \odot \phi'(z) + \lambda w^{(1)}$$

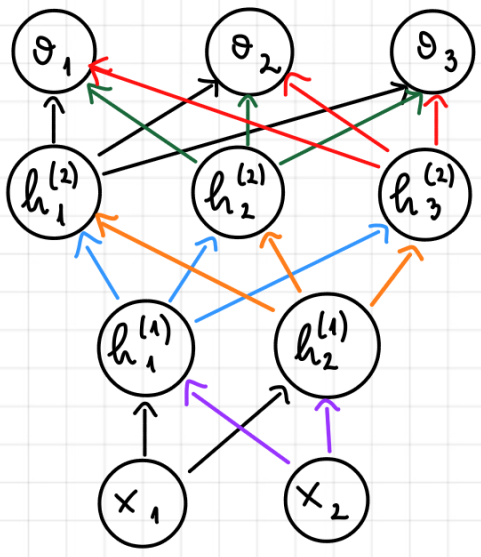


3.

5. Consider a dataset containing [illegible] average linkage to perform hierarchical clustering of the points using [illegible]

6. Draw the diagram of an MLP for regression, having 2 inputs, two hidden layers of 2 and 3 units, respectively, and 3 outputs. The activation function for both hidden layers is ReLU. For the input $X = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$, if the weight matrices and bias vectors of the layers are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the output O of the network. Specify the total number of real parameters (weights and biases) for the network.

Diagram



$$X = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \in \mathbb{R}^{n \times d}$$

$n = 2 \quad d = 2 \quad h^{(1)} = 2 \quad h^{(2)} = 3 \quad q = 3$

$$\text{nr param} = d \cdot h^{(1)} + 1 \cdot h^{(1)} + h^{(1)} \cdot h^{(2)} + 1 \cdot h^{(2)} + h^{(2)} \cdot q + 1 \cdot q =$$

$$= 27$$

Formule:

$$H_1 = \nabla(X W^{(1)} + b^{(1)})$$

$$H_2 = \nabla(H_1 W^{(2)} + b^{(2)})$$

$$O = H_2 W^{(3)} + b^{(3)}$$

$$W^{(1)} \in \mathbb{R}^{d \cdot h^{(1)} = 2 \times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad b^{(1)} \in \mathbb{R}^{1 \times h^{(1)} = 1 \cdot 2} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$W^{(2)} \in \mathbb{R}^{h^{(1)} \cdot h^{(2)} = 2 \times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad b^{(2)} \in \mathbb{R}^{1 \times h^{(2)} = 1 \cdot 3} = \begin{pmatrix} -1 & 0 & 0 \end{pmatrix}$$

$$W^{(3)} \in \mathbb{R}^{h^{(2)} \cdot q = 3 \cdot 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad b^{(3)} \in \mathbb{R}^{1 \cdot q = 1 \cdot 3} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$$

$$\sigma(x) = \text{ReLU}(x) = \max(x, 0)$$

$$H_1 = \sigma\Big(\overbrace{X\, W^{(1)} + b^{(1)}}^{A}\Big)$$

$$A = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & 0 \\ 2 & -1 \end{pmatrix}$$

$$\Rightarrow H_1 = \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix}$$

$$H_2 = \sigma\Big(\overbrace{H_1\, W^{(2)} + b^{(2)}}^{B}\Big)$$

$$B = \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow H_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$O = H_2\, W^{(3)} + b^{(3)} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + (1\ 0\ 0)$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$$

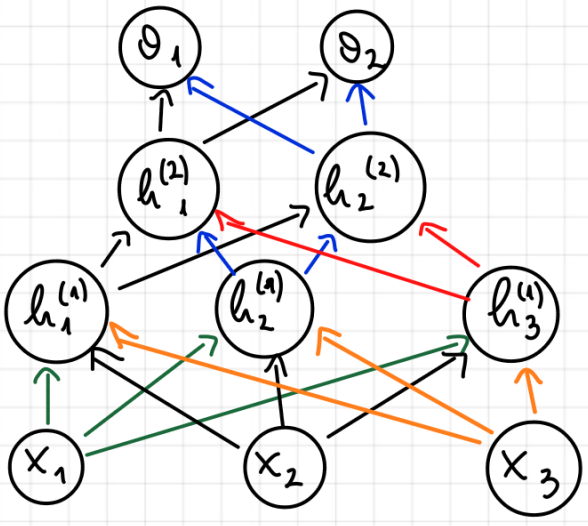
$$\in \mathbb{R}^{2\times 3}$$

4.

6. Draw the diagram of an MLP for classification, having 3 inputs, two hidden layers of 3 and 2 units, respectively, and 2 outputs. The activation function for both hidden layers is ReLU. For the input $X = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$, if the weight matrices and bias vectors of the layers are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the output O of the network. Specify the total number of real parameters (weights and biases) for the network.

ReLu

ReLu



$$x = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

$n = 2 \quad d = 3 \quad h^{(1)} = 3 \quad h^{(2)} = 2 \quad q = 2$

$W^{(1)} \in \mathbb{R}^{d \times h^{(1)}} = \mathbb{R}^{3\times 3} \rightarrow 3_3 \qquad b^{(1)} \in \mathbb{R}^{1 \times h^{(1)}} \rightarrow (1 \quad 0 \quad 0)$

$W^{(2)} \in \mathbb{R}^{h^{(1)} \times h^{(2)}}_{3\times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad b^{(2)} \in \mathbb{R}^{1\times 2} \rightarrow (1 \quad 0)$

$W^{(3)} \in \mathbb{R}^{h^{(2)} \times q}_{2\times 2} \rightarrow 3_2 \qquad b^{(3)} \in \mathbb{R}^{1\times q} = \mathbb{R}^{1\times 2} \rightarrow (1 \quad 0)$

Formule: $H_1 = \nabla(x W^{(1)} + b^{(1)})$

$H_2 = \nabla(H_1 W^{(2)} + b^{(2)})$

$O = H_2 W^{(3)} + b^{(3)}$

total nr. parametri $= 3\cdot 3 + 1\cdot 3 + 3\cdot 2 + 1\cdot 2 + 2\cdot 2 + 1\cdot 2 = 9+3+6+2+4+2 =$

$= 12 + 8 + 6 = 26$

$$H_1 = \sigma\left(\underset{2\times 3}{\begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}} \cdot \underset{3\times 3}{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}} + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \right) =$$

$$= \sigma\left(\begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \right) = \sigma\left(\begin{pmatrix} 2 & 0 & 1 \\ 2 & -1 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} 2 & 0 & 1 \\ 2 & 0 & 0 \end{pmatrix}$$

$$H_2 = \sigma\left(\begin{pmatrix} 2 & 0 & 1 \\ 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \end{pmatrix} \right) =$$

$$= \sigma\left(\begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \right) = \sigma\left(\begin{pmatrix} 3 & 0 \\ 3 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} 3 & 0 \\ 3 & 0 \end{pmatrix}$$

$$O = \begin{pmatrix} 3 & 0 \\ 3 & 0 \end{pmatrix} J_2 + \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 4 & 0 \end{pmatrix}$$